First-Order Differential Equations

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Notes with worked examples: analytical methods (separable, linear, exact), direction fields, numerical methods (Euler, Heun, RK4), and applications.

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Runge–Kutta 4 — derivations and worked examples 7. Error, order and efficiency (including Richardson error estimates) 8. Stability and stiff equations 9. Numerical methods in practice: adaptive step size, choosing a method, pitfalls 10. Applications: Newton's law of cooling, logistic growth 11. An equation with no elementary solution 12. Summary 13. Exercises

```
import numpy as np
import matplotlib.pyplot as plt

plt.rcParams["figure.figsize"] = (7, 4.5)
plt.rcParams["axes.grid"] = True
np.set_printoptions(precision=6, suppress=True)
```

1. Definitions

A **first-order ordinary differential equation (ODE)** relates an unknown function $y(x)$ to its first derivative:

$$\frac{dy}{dx} = f(x, y).$$

An **initial value problem (IVP)** adds a starting condition:

$$\frac{dy}{dx} = f(x, y), \qquad y(x_0) = y_0.$$

- A **general solution** contains an arbitrary constant C (a one-parameter family of curves).
- A **particular solution** is the member of that family that satisfies the initial condition.

Existence and uniqueness (Picard's theorem). If f and $\partial f/\partial y$ are continuous in a rectangle containing (x_0, y_0), the IVP has exactly one solution on some interval around x_0.

Example where uniqueness fails: $y' = 3y^{2/3}$, $y(0) = 0$. Both $y = 0$ and $y = x^3$ are solutions, because $\partial f/\partial y = 2y^{-1/3}$ is not continuous at $y = 0$.

2. Direction fields

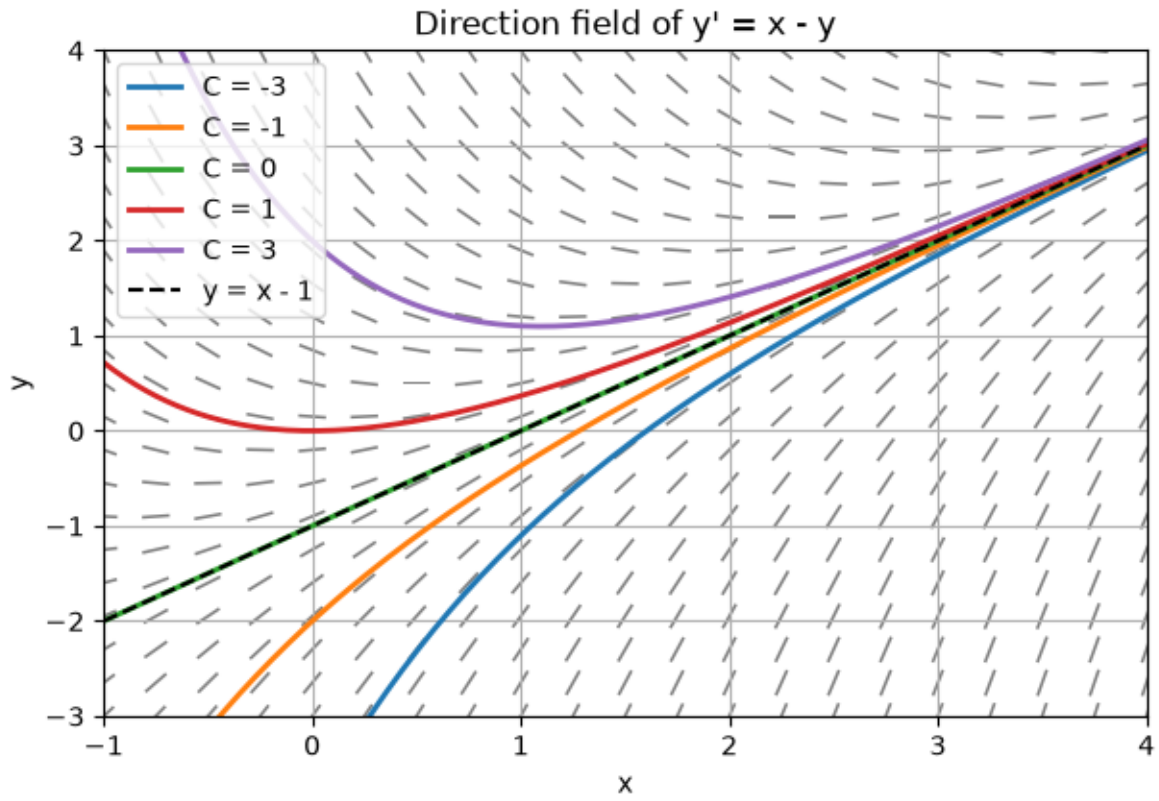
At every point (x, y) the equation gives the slope $f(x, y)$ of the solution curve passing through it. Drawing short line segments with these slopes produces a **direction (slope) field**. Solution curves follow the arrows.

Below is the field for $y' = x - y$, with a few solution curves. The exact solution is $y = x - 1 + Ce^{-x}$, so every curve approaches the line $y = x - 1$.

```
def direction_field(f, xlim, ylim, n=21, ax=None):
    # Draw normalised slope segments of y' = f(x, y).
    ax = ax or plt.gca()
    X, Y = np.meshgrid(np.linspace(*xlim, n), np.linspace(*ylim, n))
    S = f(X, Y)
    L = np.sqrt(1 + S**2)
    ax.quiver(X, Y, 1 / L, S / L, angles="xy", color="gray",
              headwidth=0, headlength=0, headaxislength=0, pivot="middle")
    ax.set_xlim(xlim); ax.set_ylim(ylim)
    ax.set_xlabel("x"); ax.set_ylabel("y")
    return ax

f = lambda x, y: x - y
ax = direction_field(f, (-1, 4), (-3, 4))
x = np.linspace(-1, 4, 200)
```

```
for C in [-3, -1, 0, 1, 3]:
    ax.plot(x, x - 1 + C * np.exp(-x), lw=2, label=f"C = {C}")
ax.plot(x, x - 1, "k--", label="y = x - 1")
ax.set_title("Direction field of y' = x - y")
ax.legend(loc="upper left");
```



3. Separable equations

An equation is **separable** if it can be written as

$$\frac{dy}{dx} = g(x)\,h(y).$$

Method: separate the variables and integrate both sides:

$$\int \frac{dy}{h(y)} = \int g(x)\,dx + C.$$

Worked example 3.1

Solve $\dfrac{dy}{dx} = xy$, $y(0) = 1$.

1. Separate: $\dfrac{dy}{y} = x\,dx$.
2. Integrate: $\ln|y| = \dfrac{x^2}{2} + C$.
3. Solve for y: $y = Ae^{x^2/2}$.

4. Apply $y(0)=1$: $A=1$, so $\boxed{y = e^{x^2/2}}$.

Check: $y' = xe^{x^2/2} = xy$. Numerically, $y(1) = e^{0.5} \approx 1.648721$.

Worked example 3.2

Solve $\dfrac{dy}{dx} = \dfrac{x^2}{1+y}$, $y(0)=0$.

1. $(1+y)\,dy = x^2\,dx$
2. $y + \dfrac{y^2}{2} = \dfrac{x^3}{3} + C$, and $y(0)=0$ gives $C=0$.
3. Solve the quadratic $y^2 + 2y - \frac{2}{3}x^3 = 0$ and take the root through the origin: $\boxed{y = -1 + \sqrt{1 + \frac{2}{3}x^3}}$.

At $x=1$: $y = -1 + \sqrt{5/3} \approx 0.290994$.

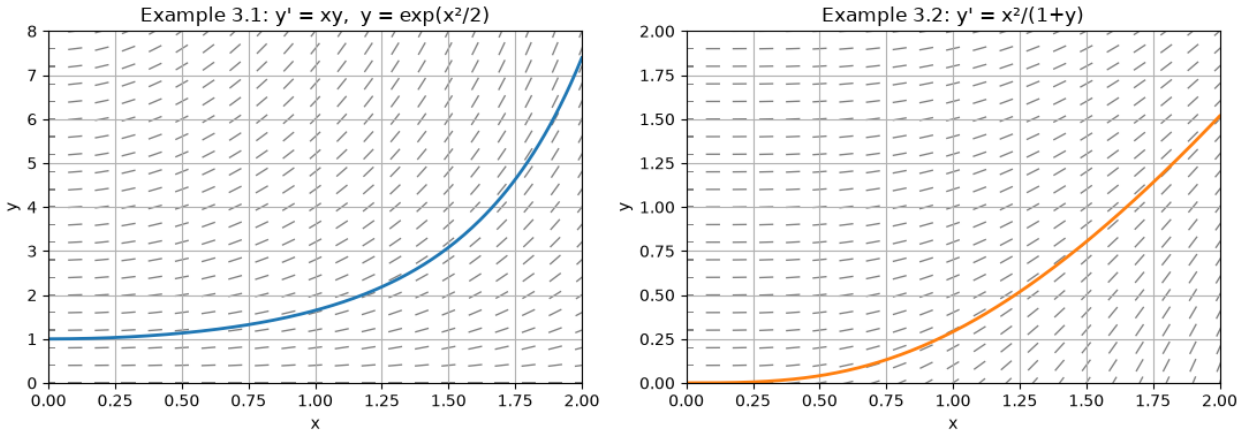
```
x = np.linspace(0, 2, 200)
fig, axes = plt.subplots(1, 2, figsize=(11, 4))

direction_field(lambda x, y: x * y, (0, 2), (0, 8), ax=axes[0])
axes[0].plot(x, np.exp(x**2 / 2), "C0", lw=2)
axes[0].set_title("Example 3.1: y' = xy,  y = exp(x²/2)")

direction_field(lambda x, y: x**2 / (1 + y), (0, 2), (0, 2), ax=axes[1])
axes[1].plot(x, -1 + np.sqrt(1 + 2 * x**3 / 3), "C1", lw=2)
axes[1].set_title("Example 3.2: y' = x²/(1+y)")
plt.tight_layout()

print("Example 3.1  y(1) =", np.exp(0.5))
print("Example 3.2  y(1) =", -1 + np.sqrt(5 / 3))
```

```
Example 3.1  y(1) = 1.6487212707001282
Example 3.2  y(1) = 0.2909944487358056
```



4. Linear first-order equations

Standard form:

$$\frac{dy}{dx} + P(x)\,y = Q(x).$$

Method (integrating factor):

1. Compute $\mu(x) = e^{\int P(x)\,dx}$.
2. Multiply through: $\mu y' + \mu P y = \mu Q$. The left side is exactly $(\mu y)'$.
3. Integrate: $\mu(x)\,y = \int \mu(x) Q(x)\,dx + C$.
4. Divide by μ.

Worked example 4.1

Solve $y' + y = x$, $y(0) = 1$.

1. $P = 1$, so $\mu = e^x$.
2. $(e^x y)' = x e^x$.
3. Integrate by parts: $e^x y = (x-1)e^x + C$.
4. $y = x - 1 + Ce^{-x}$. From $y(0) = 1$: $-1 + C = 1$, so $C = 2$.

$$\boxed{y = x - 1 + 2e^{-x}}$$

Worked example 4.2

Solve $xy' + 2y = 4x^2$, $y(1) = 2$, for $x > 0$.

1. Standard form: $y' + \dfrac{2}{x} y = 4x$, so $\mu = e^{2\ln x} = x^2$.
2. $(x^2 y)' = 4x^3$, so $x^2 y = x^4 + C$.
3. $y = x^2 + C/x^2$. From $y(1) = 2$: $C = 1$.

$$\boxed{y = x^2 + \frac{1}{x^2}}$$

```
xs = np.array([0.0, 0.5, 1.0, 1.5, 2.0])
print("Example 4.1:  y = x - 1 + 2e^(-x)")
print(f"{'x':>6} {'y(x)':>12}")
for xi in xs:
    print(f"{xi:6.2f} {xi - 1 + 2 * np.exp(-xi):12.6f}")

xs = np.array([1.0, 1.5, 2.0, 3.0])
print("\nExample 4.2:  y = x^2 + 1/x^2")
print(f"{'x':>6} {'y(x)':>12}")
for xi in xs:
    print(f"{xi:6.2f} {xi**2 + 1 / xi**2:12.6f}")
```

```
Example 4.1:  y = x - 1 + 2e^(-x)
     x         y(x)
  0.00     1.000000
  0.50     0.713061
  1.00     0.735759
  1.50     0.946260
  2.00     1.270671

Example 4.2:  y = x^2 + 1/x^2
     x         y(x)
  1.00     2.000000
  1.50     2.694444
  2.00     4.250000
  3.00     9.111111
```

5. Exact equations

An equation $M(x,y)\,dx + N(x,y)\,dy = 0$ is **exact** if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

Then there is a potential function $F(x,y)$ with $F_x = M$ and $F_y = N$, and the solution is $F(x,y) = C$.

Worked example 5.1

Solve $(2xy+3)\,dx + (x^2-1)\,dy = 0$, $y(0) = 2$.

1. Test: $M_y = 2x$, $N_x = 2x$. The equation is exact.
2. Integrate M with respect to x: $F = x^2y + 3x + g(y)$.
3. Differentiate with respect to y and match N: $x^2 + g'(y) = x^2 - 1$, so $g(y) = -y$.
4. Solution: $x^2y + 3x - y = C$. From $y(0) = 2$: $C = -2$.

$$\boxed{y = \frac{-2-3x}{x^2-1} = \frac{3x+2}{1-x^2}}, \qquad -1 < x < 1.$$

At $x = 0.5$: $y = 3.5/0.75 \approx 4.666667$.

```
# Check example 5.1 by verifying that F(x, y(x)) stays equal to C = -2
y = lambda x: (3 * x + 2) / (1 - x**2)
F = lambda x, y: x**2 * y + 3 * x - y
for xi in [0.0, 0.25, 0.5, 0.75]:
    print(f"x = {xi:4.2f}   y = {y(xi):9.6f}   F(x, y) = {F(xi, y(xi)):.6f}")
```

```
x = 0.00   y =  2.000000   F(x, y) = -2.000000
x = 0.25   y =  2.933333   F(x, y) = -2.000000
x = 0.50   y =  4.666667   F(x, y) = -2.000000
x = 0.75   y =  9.714286   F(x, y) = -2.000000
```

6. Numerical methods

6.1 The idea: replace the curve by a sequence of points

Most ODEs have no closed-form solution, but we can still compute one approximately. Instead of finding a *function* $y(x)$, a numerical method computes *numbers* $y_0, y_1, y_2, \ldots$ that approximate y on a grid

$$x_n = x_0 + nh, \qquad n = 0, 1, 2, \ldots$$

where h is the **step size**. Every method in this notebook is a **one-step method**: it uses only the current point (x_n, y_n) and the slope function f to produce the next point y_{n+1}.

The starting point for all of them is the ODE written in integral form. Integrating $y' = f(x,y)$ from x_n to x_{n+1} gives the **exact** relation

$$y(x_{n+1}) = y(x_n) + \int_{x_n}^{x_{n+1}} f(x, y(x))\,dx.$$

The integral is the only part we cannot compute exactly, because $y(x)$ inside it is unknown. **Each method is a different way of approximating that integral**, that is, of estimating the *average slope* over one step:

$$y_{n+1} = y_n + h \times (\text{estimate of the average slope on } [x_n, x_{n+1}]).$$

Method	Average slope estimated by	Like the quadrature rule
Euler	slope at the left end	left rectangle rule
Heun	mean of the slopes at both ends	trapezoidal rule
RK4	weighted mean of slopes at the start, midpoint (twice) and end	Simpson's rule

Throughout this section we use one **test problem** whose exact solution is known, so every error can be measured:

$$y' = x + y, \qquad y(0) = 1, \qquad \text{exact solution } y = 2e^x - x - 1.$$

```
f = lambda x, y: x + y
exact = lambda x: 2 * np.exp(x) - x - 1
```

6.2 Euler's method

Derivation from the Taylor series. Expand the exact solution about x_n:

$$y(x_n + h) = y(x_n) + h\,y'(x_n) + \frac{h^2}{2}\,y''(\xi), \qquad \xi \in (x_n, x_{n+1}).$$

Since $y'(x_n) = f(x_n, y_n)$, dropping the h^2 term gives **Euler's method**:

$$\boxed{y_{n+1} = y_n + h\,f(x_n, y_n)}$$

Geometric meaning. From the current point, walk along the tangent line for a distance h. Because the solution curve bends away from its tangent, each step lands slightly off the curve. The next step then starts from the wrong point, on a *neighbouring* solution curve, so the errors build up.

Algorithm 1. Start with x_0, y_0, step size h and number of steps N. 2. For $n = 0, 1, \dots, N-1$: * compute the slope $k = f(x_n, y_n)$ * set $y_{n+1} = y_n + hk$ and $x_{n+1} = x_n + h$.

```
def euler(f, x0, y0, h, n):
    x, y = np.zeros(n + 1), np.zeros(n + 1)
    x[0], y[0] = x0, y0
    for i in range(n):
        y[i + 1] = y[i] + h * f(x[i], y[i])
        x[i + 1] = x[i] + h
    return x, y
```

Worked example: Euler by hand, $h = 0.1$

n	x_n	y_n	slope $f(x_n, y_n) = x_n + y_n$	$y_{n+1} = y_n + 0.1 \times \text{slope}$	exact $y(x_{n+1})$	error
0	0.0	1.000000	1.000000	1.100000	1.110342	0.0103
1	0.1	1.100000	1.200000	1.220000	1.242806	0.0228
2	0.2	1.220000	1.420000	1.362000	1.399718	0.0377

n	x_n	y_n	slope $f(x_n, y_n) = x_n + y_n$	$y_{n+1} = y_n + 0.1 \times \text{slope}$	exact $y(x_{n+1})$	error

The error grows every step, because each step starts from an already inaccurate value **and** adds a new error of its own.

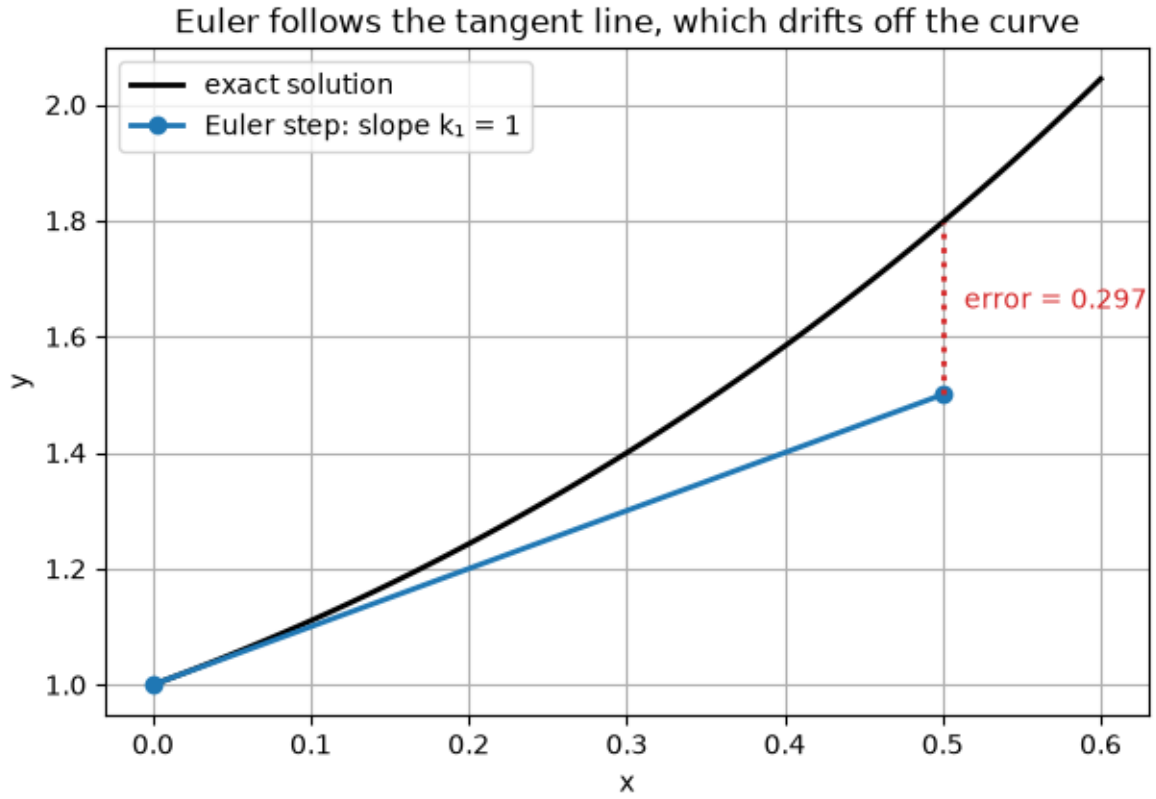
Local and global error

- The **local truncation error** (LTE) is the error made in *one* step, starting from the exact value. It is the term we dropped: $\frac{h^2}{2}y''(\xi) = O(h^2)$. *Check:* here $y'' = 1 + y' = 1 + x + y$, so $y''(0) = 2$ and the predicted first-step error is $\frac{0.1^2}{2} \cdot 2 = 0.01$. The actual first-step error is 0.0103.
- The **global error** is the total error at a fixed point $x = b$ after all $N = (b - x_0)/h$ steps. Roughly N local errors of size $O(h^2)$ add up:

$$\text{global error} \approx N \cdot O(h^2) = \frac{b - x_0}{h} O(h^2) = O(h).$$

A method whose global error is $O(h^p)$ is said to have **order** p. Euler has order 1, so halving h only halves the error. Gaining one more correct digit costs 10 times as many steps.

```
# Geometric picture: one large Euler step (h = 0.5) so the gap is visible
h = 0.5
xf = np.linspace(0, 0.6, 100)
plt.plot(xf, exact(xf), "k", lw=2, label="exact solution")
k1 = f(0, 1)
plt.plot([0, h], [1, 1 + h * k1], "C0o-", lw=2, label=f"Euler step: slope k  = {k1:g}")
plt.vlines(h, 1 + h * k1, exact(h), colors="C3", linestyles=":", lw=2)
plt.annotate(f"error = {exact(h) - (1 + h*k1):.3f}", (h, (1 + h*k1 + exact(h)) / 2),
             xytext=(8, 0), textcoords="offset points", color="C3")
plt.title("Euler follows the tangent line, which drifts off the curve")
plt.xlabel("x"); plt.ylabel("y"); plt.legend();
```



6.3 Heun's method (improved Euler)

Derivation from the trapezoidal rule. Estimate the integral with the trapezoidal rule, which averages the slopes at both ends of the step:

$$y_{n+1} = y_n + \frac{h}{2}\Big[f(x_n, y_n) + f(x_{n+1}, y_{n+1})\Big].$$

The problem is that y_{n+1} appears on both sides (the formula is **implicit**). Heun's fix is to first *predict* y_{n+1} with an Euler step and then *correct* it with the trapezoidal average. That makes it a **predictor–corrector** method:

$$\begin{aligned} k_1 &= f(x_n, y_n) && \text{slope at the start} \\ y^*_{n+1} &= y_n + hk_1 && \text{predictor (an Euler step)} \\ k_2 &= f(x_{n+1}, y^*_{n+1}) && \text{slope at the predicted end point} \\ y_{n+1} &= y_n + \tfrac{h}{2}(k_1 + k_2) && \text{corrector: use the average slope} \end{aligned}$$

Why it is better. If the curve is concave up, the start slope k_1 is too small and the end slope k_2 is too large, so their average largely cancels the error. The local error becomes $O(h^3)$ and the global error $O(h^2)$, so Heun is **order 2**. Halving h divides the error by about 4. Each step costs **2** evaluations of f.

```
def heun(f, x0, y0, h, n):
    x, y = np.zeros(n + 1), np.zeros(n + 1)
    x[0], y[0] = x0, y0
    for i in range(n):
        k1 = f(x[i], y[i])
        k2 = f(x[i] + h, y[i] + h * k1)
```

```
        y[i + 1] = y[i] + h / 2 * (k1 + k2)
        x[i + 1] = x[i] + h
    return x, y
```

Worked example: Heun by hand, $h = 0.1$

n	x_n	y_n	$k_1 = x_n + y_n$	$y^* = y_n + 0.1k_1$	$k_2 = x_{n+1} + y^*$	$y_{n+1} = y_n + 0.05(k_1 + k_2)$	error
0	0.0	1.000000	1.000000	1.100000	1.200000	1.110000	0.00034
1	0.1	1.110000	1.210000	1.231000	1.431000	1.242050	0.00076
2	0.2	1.242050	1.442050	1.386255	1.686255	1.398465	0.00125

After three steps the error is 0.00125, about **30 times smaller** than Euler's 0.0377, for twice the work.

6.4 The classical Runge–Kutta method (RK4)

Derivation from Simpson's rule. Simpson's rule weights the midpoint four times as heavily as the ends:

$$\int_{x_n}^{x_{n+1}} f\,dx \approx \frac{h}{6}\Big[f_{\text{start}} + 4f_{\text{mid}} + f_{\text{end}}\Big].$$

RK4 estimates the midpoint slope **twice**, each time more accurately than the last, and splits the weight 4 into $2 + 2$:

$$\begin{aligned}
k_1 &= f(x_n,\ y_n) && \text{slope at the start}\\
k_2 &= f\big(x_n + \tfrac{h}{2},\ y_n + \tfrac{h}{2}k_1\big) && \text{slope at the midpoint, reached using } k_1\\
k_3 &= f\big(x_n + \tfrac{h}{2},\ y_n + \tfrac{h}{2}k_2\big) && \text{slope at the midpoint, reached using the better } k_2\\
k_4 &= f(x_n + h,\ y_n + hk_3) && \text{slope at the end, reached using } k_3\\
y_{n+1} &= y_n + \tfrac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)
\end{aligned}$$

The weights $\frac{1}{6}, \frac{2}{6}, \frac{2}{6}, \frac{1}{6}$ add up to 1, so $\frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$ is a weighted *average slope*. The local error is $O(h^5)$ and the global error $O(h^4)$: RK4 is **order 4**, and halving h divides the error by about 16. Each step costs **4** evaluations of f.

```
def rk4(f, x0, y0, h, n):
    x, y = np.zeros(n + 1), np.zeros(n + 1)
    x[0], y[0] = x0, y0
    for i in range(n):
        k1 = f(x[i], y[i])
        k2 = f(x[i] + h / 2, y[i] + h / 2 * k1)
        k3 = f(x[i] + h / 2, y[i] + h / 2 * k2)
        k4 = f(x[i] + h, y[i] + h * k3)
        y[i + 1] = y[i] + h / 6 * (k1 + 2 * k2 + 2 * k3 + k4)
        x[i + 1] = x[i] + h
    return x, y
```

Worked example: RK4 by hand, first step with $h = 0.1$

stage	evaluation point (x, y)	slope
k_1	$(0,\ 1)$	1
k_2	$(0.05,\ 1 + 0.05 \cdot 1) = (0.05,\ 1.05)$	1.1
k_3	$(0.05,\ 1 + 0.05 \cdot 1.1) = (0.05,\ 1.055)$	1.105
k_4	$(0.1,\ 1 + 0.1 \cdot 1.105) = (0.1,\ 1.1105)$	1.2105

$$y_1 = 1 + \frac{0.1}{6}(1 + 2(1.1) + 2(1.105) + 1.2105) = 1 + \frac{0.1}{6}(6.6205) = 1.1103417.$$

Exact: $y(0.1) = 1.1103418$. The error is 1.7×10^{-7}, about 60,000 times smaller than Euler's first step.

6.5 Seeing the slopes

Below, one large step ($h = 0.5$) from $(0, 1)$ shows where each method samples the slope. Short arrows mark each sampled slope; the coloured line is the step actually taken.

```
h, x0, y0 = 0.5, 0.0, 1.0
xf = np.linspace(0, 0.55, 100)

def slope_mark(ax, x, y, s, color, label=None):
    d = 0.06
    ax.plot([x - d, x + d], [y - d * s, y + d * s], color=color, lw=3, label=label)
    ax.plot(x, y, "o", color=color, ms=5)

fig, axes = plt.subplots(1, 3, figsize=(14, 4.2), sharey=True)
for ax in axes:
    ax.plot(xf, exact(xf), "k", lw=2, label="exact")
    ax.set_xlabel("x")
axes[0].set_ylabel("y")

# Euler
k1 = f(x0, y0)
y_e = y0 + h * k1
slope_mark(axes[0], x0, y0, k1, "C0", "k ")
axes[0].plot([x0, h], [y0, y_e], "C0--")
axes[0].set_title(f"Euler: 1 slope\ny  = {y_e:.4f}, error {exact(h) - y_e:.4f}")

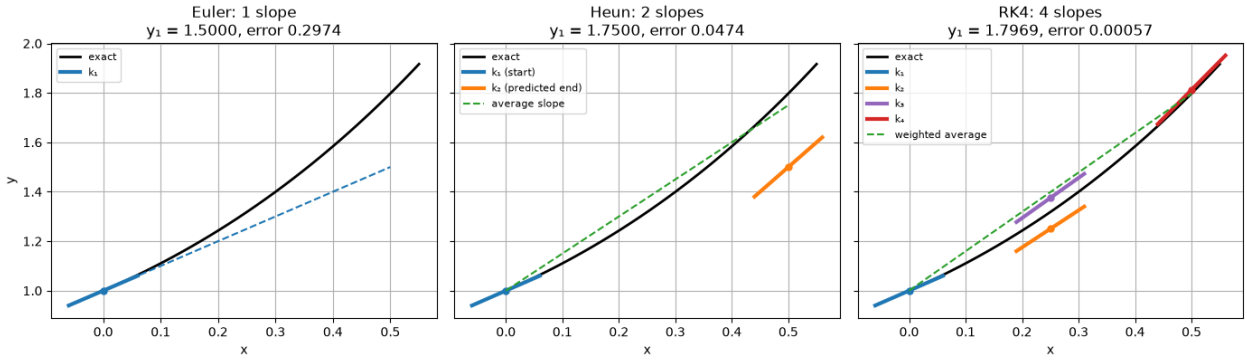
# Heun
k2 = f(x0 + h, y0 + h * k1)
y_h = y0 + h / 2 * (k1 + k2)
slope_mark(axes[1], x0, y0, k1, "C0", "k  (start)")
slope_mark(axes[1], x0 + h, y0 + h * k1, k2, "C1", "k  (predicted end)")
axes[1].plot([x0, h], [y0, y_h], "C2--", label="average slope")
axes[1].set_title(f"Heun: 2 slopes\ny  = {y_h:.4f}, error {exact(h) - y_h:.4f}")

# RK4
r1 = f(x0, y0)
r2 = f(x0 + h / 2, y0 + h / 2 * r1)
r3 = f(x0 + h / 2, y0 + h / 2 * r2)
r4 = f(x0 + h, y0 + h * r3)
y_r = y0 + h / 6 * (r1 + 2 * r2 + 2 * r3 + r4)
slope_mark(axes[2], x0, y0, r1, "C0", "k ")
slope_mark(axes[2], x0 + h / 2, y0 + h / 2 * r1, r2, "C1", "k ")
```

```
slope_mark(axes[2], x0 + h / 2, y0 + h / 2 * r2, r3, "C4", "k ")
slope_mark(axes[2], x0 + h, y0 + h * r3, r4, "C3", "k ")
axes[2].plot([x0, h], [y0, y_r], "C2--", label="weighted average")
axes[2].set_title(f"RK4: 4 slopes\ny  = {y_r:.4f}, error {exact(h) - y_r:.5f}")

for ax in axes:
    ax.legend(loc="upper left", fontsize=8)
plt.tight_layout()
print(f"exact y(0.5) = {exact(h):.6f}")
```

```
exact y(0.5) = 1.797443
```



6.6 Why the order is what it is: matching the Taylor series

For the test problem, the exact solution after one step from $(0, 1)$ has the Taylor series

$$y(h) = 2e^h - h - 1 = 1 + h + h^2 + \frac{h^3}{3} + \frac{h^4}{12} + \frac{h^5}{60} + \cdots$$

Expanding each method's formula for one step, symbolically in h:

Method	One step gives	First term missed (local error)	Predicted at $h = 0.1$	Actual
Euler	$1 + h$	h^2	1.0×10^{-2}	1.03×10^{-2}
Heun	$1 + h + h^2$	$h^3/3$	3.3×10^{-4}	3.42×10^{-4}
RK4	$1 + h + h^2 + \frac{h^3}{3} + \frac{h^4}{12}$	$h^5/60$	1.7×10^{-7}	1.69×10^{-7}

(For RK4: $k_1 = 1$, $k_2 = 1 + h$, $k_3 = 1 + h + \frac{h^2}{2}$, $k_4 = 1 + 2h + h^2 + \frac{h^3}{2}$; substituting gives the expression in the table.)

This is the real meaning of order: **a method of order p reproduces the Taylor series of the true solution through the h^p term.** The first term it misses, of size h^{p+1}, is the local error. Over $1/h$ steps these add up to a global error of h^p.

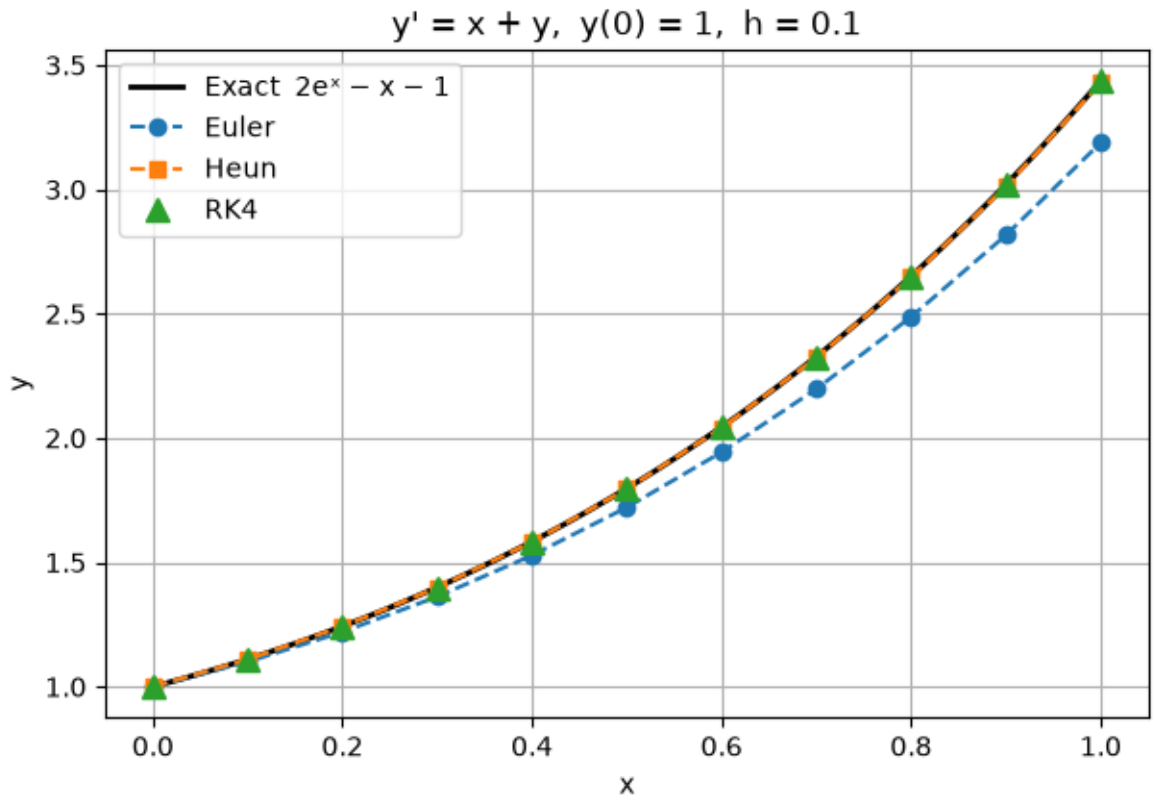
6.7 All three methods side by side

```
h, n = 0.1, 10
x, ye = euler(f, 0, 1, h, n)
_, yh = heun(f, 0, 1, h, n)
_, yr = rk4(f, 0, 1, h, n)
yx = exact(x)
```

```
print(f"{'x':>5} {'Euler':>10} {'Heun':>10} {'RK4':>12} {'Exact':>12} "
      f"{'|err| Eul':>10} {'|err| Heun':>10} {'|err| RK4':>10}")
for i in range(n + 1):
    print(f"{x[i]:5.1f} {ye[i]:10.6f} {yh[i]:10.6f} {yr[i]:12.8f} {yx[i]:12.8f} "
          f"{abs(ye[i]-yx[i]):10.2e} {abs(yh[i]-yx[i]):10.2e} {abs(yr[i]-yx[i]):10.2e}")
```

```
    x      Euler       Heun          RK4        Exact  |err| Eul |err| Heun  |err| RK4
  0.0   1.000000   1.000000   1.00000000   1.00000000   0.00e+00   0.00e+00   0.00e+00
  0.1   1.100000   1.110000   1.11034167   1.11034184   1.03e-02   3.42e-04   1.69e-07
  0.2   1.220000   1.242050   1.24280514   1.24280552   2.28e-02   7.56e-04   3.75e-07
  0.3   1.362000   1.398465   1.39971699   1.39971762   3.77e-02   1.25e-03   6.21e-07
  0.4   1.528200   1.581804   1.58364848   1.58364940   5.54e-02   1.85e-03   9.15e-07
  0.5   1.721020   1.794894   1.79744128   1.79744254   7.64e-02   2.55e-03   1.26e-06
  0.6   1.943122   2.040857   2.04423592   2.04423760   1.01e-01   3.38e-03   1.68e-06
  0.7   2.197434   2.323147   2.32750325   2.32750541   1.30e-01   4.36e-03   2.16e-06
  0.8   2.487178   2.645578   2.65107913   2.65108186   1.64e-01   5.50e-03   2.73e-06
  0.9   2.815895   3.012364   3.01920283   3.01920622   2.03e-01   6.84e-03   3.39e-06
  1.0   3.187485   3.428162   3.43655949   3.43656366   2.49e-01   8.40e-03   4.17e-06
```

```
xf = np.linspace(0, 1, 200)
plt.plot(xf, exact(xf), "k", lw=2, label="Exact  2e  - x - 1")
plt.plot(x, ye, "o--", label="Euler")
plt.plot(x, yh, "s--", label="Heun")
plt.plot(x, yr, "^", ms=8, label="RK4")
plt.title("y' = x + y,  y(0) = 1,  h = 0.1")
plt.xlabel("x"); plt.ylabel("y"); plt.legend();
```



7. Error, order and efficiency

7.1 Measuring the order

If the global error is $E(h) \approx Ch^p$, then

$$\frac{E(h)}{E(h/2)} \approx 2^p, \qquad \log E \approx \log C + p \log h.$$

So the order can be read off in two ways: from the **error ratio** when h is halved, or from the **slope of a log–log plot** of error against h.

Method	Order p	Error ratio when h is halved
Euler	1	2
Heun	2	4
RK4	4	16

We check this for the test problem by measuring the error at $x = 1$ (exact value $2e - 2 \approx 3.436564$).

```
hs = [0.2, 0.1, 0.05, 0.025, 0.0125]
methods = {"Euler": euler, "Heun": heun, "RK4": rk4}
cost_per_step = {"Euler": 1, "Heun": 2, "RK4": 4}
errors = {name: [] for name in methods}

for h in hs:
    n = round(1 / h)
    for name, method in methods.items():
        _, y = method(f, 0, 1, h, n)
        errors[name].append(abs(y[-1] - exact(1.0)))

print(f"{'h':>8}" + "".join(f"{m:>14}{'ratio':>8}" for m in methods))
for i, h in enumerate(hs):
    row = f"{h:8.4f}"
    for name in methods:
        e = errors[name]
        ratio = f"{e[i-1] / e[i]:8.2f}" if i else f"{'':>8}"
        row += f"{e[i]:14.3e}{ratio}"
    print(row)

for name, p in zip(methods, [1, 2, 4]):
    slope = np.polyfit(np.log(hs), np.log(errors[name]), 1)[0]
    print(f"{name:6s} observed order  {slope:.2f}  (theory {p})")
```

```
       h         Euler   ratio          Heun   ratio           RK4   ratio
  0.2000     4.599e-01              3.115e-02              6.138e-05
  0.1000     2.491e-01    1.85      8.402e-03    3.71      4.169e-06   14.73
  0.0500     1.300e-01    1.92      2.182e-03    3.85      2.716e-07   15.35
  0.0250     6.644e-02    1.96      5.558e-04    3.93      1.733e-08   15.67
  0.0125     3.359e-02    1.98      1.403e-04    3.96      1.095e-09   15.83
Euler  observed order  0.95  (theory 1)
Heun   observed order  1.95  (theory 2)
RK4    observed order  3.95  (theory 4)
```

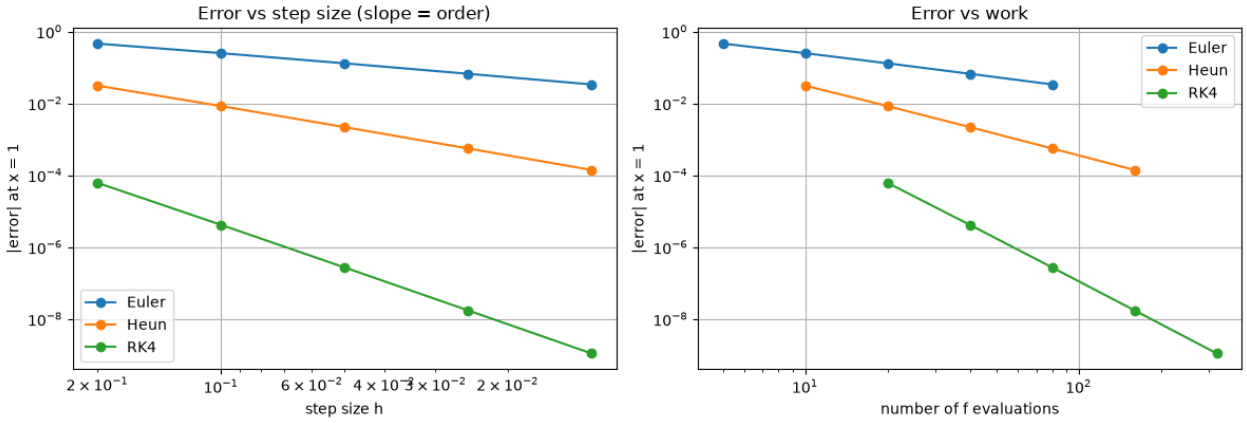
7.2 Accuracy per unit of work

Smaller steps are not free. The real cost of a method is the number of times it evaluates f: N for Euler, $2N$ for Heun and $4N$ for RK4. A fair comparison plots the error against that cost.

```
fig, axes = plt.subplots(1, 2, figsize=(12, 4.2))
for name in methods:
    axes[0].loglog(hs, errors[name], "o-", label=name)
    evals = [cost_per_step[name] * round(1 / h) for h in hs]
    axes[1].loglog(evals, errors[name], "o-", label=name)
axes[0].invert_xaxis()
axes[0].set_xlabel("step size h"); axes[0].set_ylabel("|error| at x = 1")
axes[0].set_title("Error vs step size (slope = order)")
axes[1].set_xlabel("number of f evaluations"); axes[1].set_ylabel("|error| at x = 1")
axes[1].set_title("Error vs work")
for ax in axes:
    ax.legend()
plt.tight_layout()

# Work needed for an error of 1e-6, using E  C h^p from the finest run
for name, p in zip(methods, [1, 2, 4]):
    evals = cost_per_step[name] * round(1 / hs[-1])
    need = evals * (errors[name][-1] / 1e-6) ** (1 / p)
    print(f"{name:6s} needs about {need:12,.0f} evaluations of f for error 1e-6")
```

```
Euler  needs about    2,687,502 evaluations of f for error 1e-6
Heun   needs about        1,895 evaluations of f for error 1e-6
RK4    needs about           58 evaluations of f for error 1e-6
```



Takeaway. In the right-hand plot, RK4's line lies far below the others: for any fixed amount of work it is far more accurate. To reach an error of 10^{-6}, Euler needs about 2.7 million evaluations of f, Heun about 1,900 and RK4 about 60. Higher-order methods win whenever the solution is smooth.

7.3 Estimating the error when the exact solution is unknown

In real problems there is no exact solution to compare with. **Richardson's idea:** solve twice, with step sizes h and $h/2$. Since $E(h) \approx Ch^p$,

$$y(b) - y_{h/2} \approx \frac{y_{h/2} - y_h}{2^p - 1}.$$

- The right-hand side is an **error estimate** for the finer solution, computed using only numerical results.
- Adding it to $y_{h/2}$ gives an **extrapolated** value that is more accurate than either run: one order higher.

Below we check the estimate against the true error for the test problem.

```
print(f"{'method':>6} {'y_h':>12} {'y_h/2':>12} {'estimated err':>14} {'true err':>12} {'extrapolated':>
for name, p in [("Euler", 1), ("Heun", 2), ("RK4", 4)]:
    method = methods[name]
    _, y1 = method(f, 0, 1, 0.1, 10)
    _, y2 = method(f, 0, 1, 0.05, 20)
    est = (y2[-1] - y1[-1]) / (2**p - 1)
    true = exact(1.0) - y2[-1]
    extrap = y2[-1] + est
    print(f"{name:>6} {y1[-1]:12.8f} {y2[-1]:12.8f} {est:14.3e} {true:12.3e} "
          f"{extrap:13.8f} {abs(exact(1.0) - extrap):10.2e}")
```

```
method          y_h         y_h/2  estimated err     true err  extrapolated    its err
 Euler   3.18748492   3.30659541      1.191e-01    1.300e-01    3.42570590   1.09e-02
  Heun   3.42816169   3.43438211      2.073e-03    2.182e-03    3.43645558   1.08e-04
   RK4   3.43655949   3.43656339      2.598e-07    2.716e-07    3.43656365   1.18e-08
```

The estimated errors closely match the true errors, so the computed solutions **tell you how accurate they are**. Extrapolation also helps: the extrapolated Euler value is about 12 times more accurate than the finer Euler run, and the extrapolated RK4 value is accurate to about 10^{-8}. This is the standard check in practice: **always re-run with half the step size and compare.**

8. Stability: when a small error explodes

Accuracy is not the only concern. A method can also be **unstable**: for a step that is too large, errors are amplified at every step and the numerical solution blows up, even though the true solution decays.

8.1 The test equation

Take $y' = -\lambda y$ with $\lambda > 0$, $y(0) = 1$. The exact solution $y = e^{-\lambda x}$ decays to 0. Euler gives

$$y_{n+1} = y_n - h\lambda y_n = (1 - h\lambda)\, y_n \quad \Longrightarrow \quad y_n = (1 - h\lambda)^n.$$

$R = 1 - h\lambda$ is the **amplification factor**. The numerical solution decays only if $|R| < 1$, that is

$$\boxed{h < \frac{2}{\lambda}} \quad \text{(Euler's stability limit).}$$

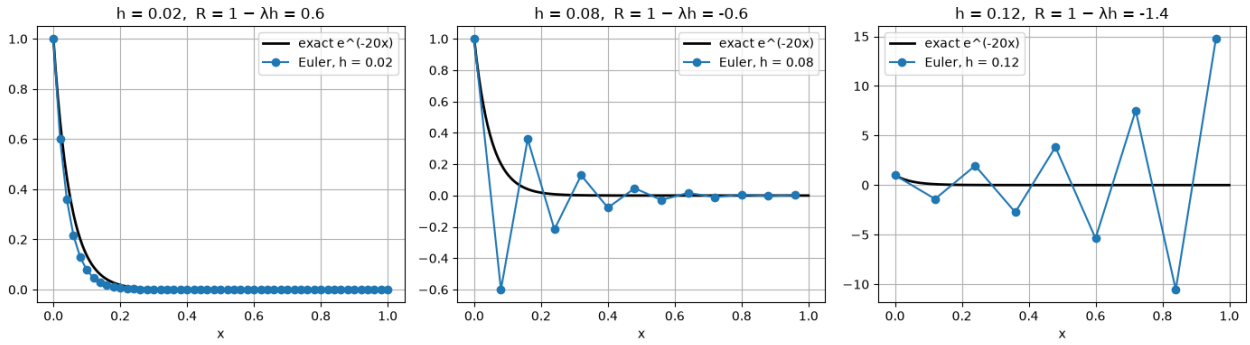
With $\lambda = 20$ the limit is $h < 0.1$:

h	$R = 1 - 20h$	behaviour
0.02	0.6	smooth decay, like the exact solution
0.08	-0.6	decays, but flips sign every step
0.12	-1.4	oscillates and grows without bound

```
lam = 20.0
decay = lambda x, y: -lam * y
xf = np.linspace(0, 1, 300)

fig, axes = plt.subplots(1, 3, figsize=(14, 4), sharey=False)
for ax, h in zip(axes, [0.02, 0.08, 0.12]):
    n = round(1 / h)
```

```
    x, y = euler(decay, 0, 1, h, n)
    ax.plot(xf, np.exp(-lam * xf), "k", lw=2, label="exact e^(-20x)")
    ax.plot(x, y, "o-", label=f"Euler, h = {h}")
    ax.set_title(f"h = {h},  R = 1 − λh = {1 - lam*h:.1f}")
    ax.set_xlabel("x"); ax.legend()
plt.tight_layout()
```



8.2 Implicit methods and stiff equations

Backward (implicit) Euler uses the slope at the *end* of the step:

$$y_{n+1} = y_n + h\, f(x_{n+1}, y_{n+1}).$$

For the test equation this gives $y_{n+1} = y_n - h\lambda y_{n+1}$, so

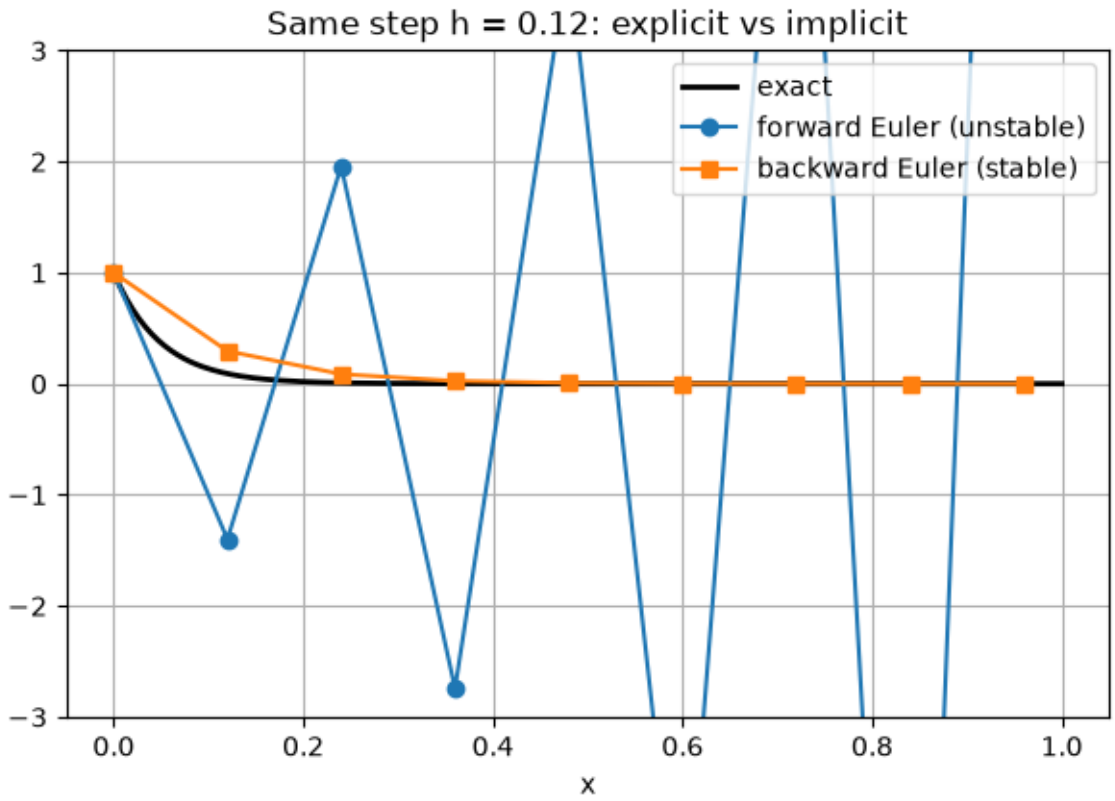
$$y_{n+1} = \frac{y_n}{1 + h\lambda}, \qquad |R| = \frac{1}{1 + h\lambda} < 1 \text{ for every } h > 0.$$

It is **unconditionally stable**. The price is that, in general, each step requires *solving* an equation for y_{n+1}, usually with Newton's method.

An equation is called **stiff** when stability, not accuracy, forces explicit methods to take tiny steps. This typically happens when the solution has a very fast-decaying part alongside a slow part. Explicit methods (Euler, Heun, RK4) all have a stability limit of the form $h < c/\lambda$; for RK4 it is $c \approx 2.79$. For stiff problems, implicit methods are much cheaper overall.

```
def backward_euler_linear(lam, y0, h, n):
    # Backward Euler for y' = -lam * y (the implicit equation solves in closed form)
    x = np.arange(n + 1) * h
    y = y0 / (1 + h * lam) ** np.arange(n + 1)
    return x, y

h = 0.12
x, y_fwd = euler(decay, 0, 1, h, round(1 / h))
_, y_bwd = backward_euler_linear(lam, 1, h, round(1 / h))
plt.plot(xf, np.exp(-lam * xf), "k", lw=2, label="exact")
plt.plot(x, y_fwd, "o-", label="forward Euler (unstable)")
plt.plot(x, y_bwd, "s-", label="backward Euler (stable)")
plt.ylim(-3, 3)
plt.title(f"Same step h = {h}: explicit vs implicit")
plt.xlabel("x"); plt.legend();
```



9. Numerical methods in practice

9.1 Adaptive step size

A fixed h wastes effort where the solution is smooth and is too coarse where it changes quickly. **Adaptive** solvers choose h step by step:

1. Take a step with two methods of different order, e.g. Euler (order 1) and Heun (order 2), which share k_1.
2. Their difference estimates the local error: $\text{err} \approx |y_{\text{Heun}} - y_{\text{Euler}}| = \frac{h}{2}|k_2 - k_1|$.
3. If $\text{err} \le \text{tol}\,(1 + |y|)$ (a mixed absolute/relative tolerance), accept the step; otherwise reject it and retry with a smaller h.
4. Choose the next step as $h_{\text{new}} = 0.9\,h\sqrt{\text{allowed}/\text{err}}$ (the square root because the local error of the lower-order method is $O(h^2)$).

Professional solvers use the same idea with higher-order pairs: **RK45 (Dormand–Prince)** in `scipy.integrate.solve_ivp`, MATLAB's `ode45`, or `Tsit5` in Julia's DifferentialEquations.jl.

Example. $y' = 1+y^2$, $y(0) = 0$, with exact solution $y = \tan x$, which becomes very steep near $x = \pi/2 \approx 1.571$. We integrate to $x = 1.5$.

```
def adaptive_heun(f, x0, y0, x_end, tol=1e-3, h=0.1):
    xs, ys = [x0], [y0]
    x, y = x0, y0
    rejected = 0
    while x_end - x > 1e-12:
        h = min(h, x_end - x)
        k1 = f(x, y)
        k2 = f(x + h, y + h * k1)
        err = h / 2 * abs(k2 - k1)
        allowed = tol * (1 + abs(y))
```

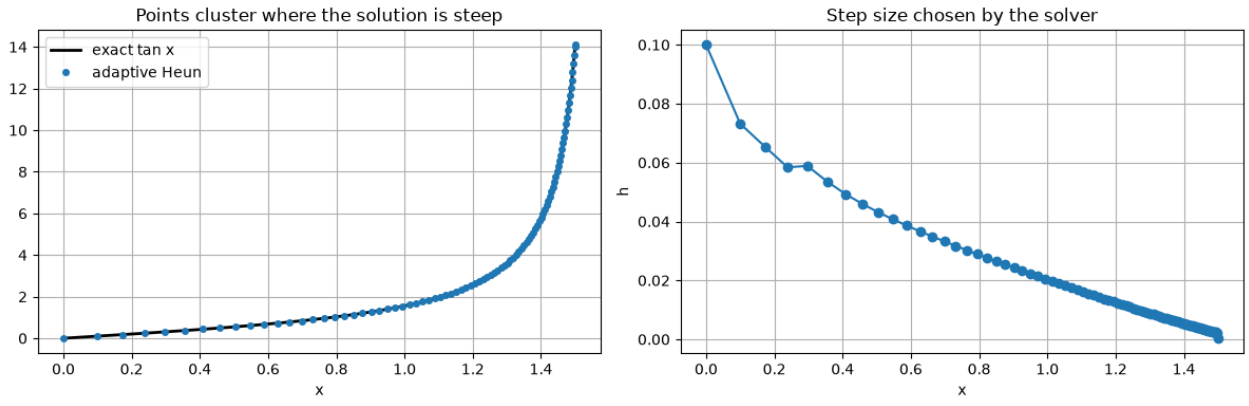
```
        if err <= allowed:
            x, y = x + h, y + h / 2 * (k1 + k2)
            xs.append(x); ys.append(y)
        else:
            rejected += 1
        h *= min(2.0, max(0.2, 0.9 * np.sqrt(allowed / err))) if err > 0 else 2.0
    return np.array(xs), np.array(ys), rejected

tanf = lambda x, y: 1 + y**2
xa, ya, rej = adaptive_heun(tanf, 0.0, 0.0, 1.5)
print(f"accepted steps: {len(xa) - 1},  rejected: {rej}")
print(f"exact tan(1.5) = {np.tan(1.5):.5f}")
print(f"adaptive Heun:              y(1.5) = {ya[-1]:.5f}   error = {abs(ya[-1] - np.tan(1.5)):.4f}")

n_fixed = len(xa) - 1
xh, yh_fixed = heun(tanf, 0, 0, 1.5 / n_fixed, n_fixed)
print(f"fixed-step Heun, {n_fixed} steps: y(1.5) = {yh_fixed[-1]:.5f}   error = {abs(yh_fixed[-1] - np.t

fig, axes = plt.subplots(1, 2, figsize=(12, 4))
xf = np.linspace(0, 1.5, 300)
axes[0].plot(xf, np.tan(xf), "k", lw=2, label="exact tan x")
axes[0].plot(xa, ya, "o", ms=4, label="adaptive Heun")
axes[0].set_title("Points cluster where the solution is steep")
axes[0].set_xlabel("x"); axes[0].legend()
axes[1].plot(xa[:-1], np.diff(xa), "o-")
axes[1].set_title("Step size chosen by the solver")
axes[1].set_xlabel("x"); axes[1].set_ylabel("h")
plt.tight_layout()
```

```
accepted steps: 89,  rejected: 3
exact tan(1.5) = 14.10142
adaptive Heun:              y(1.5) = 14.09386   error = 0.0076
fixed-step Heun, 89 steps: y(1.5) = 13.81361   error = 0.2878
```



With the same number of steps, the adaptive solver is about 40 times more accurate than fixed steps. It takes large steps where $\tan x$ is flat and puts its effort near $x = 1.5$, where the solution is steep.

9.2 Choosing a method

Situation	Good choice
Learning, or a quick rough look	Euler
Smooth, non-stiff problem	RK4 with a fixed step, or adaptive RK45
Solution changes scale a lot	Adaptive step size (RK45, Tsit5)
Stiff problem (fast and slow scales)	Implicit method: backward Euler, BDF, Rosenbrock

9.3 Common pitfalls

- **Step too large:** inaccurate, or unstable (Section 8).
- **Step too small:** wasted work, and eventually **round-off error** grows. Each step adds a tiny rounding error of about 10^{-16}, and millions of steps add those up.
- **No convergence check:** a single run tells you nothing about its accuracy. Halve h, or tighten the tolerance, and compare (Section 7.3).
- **Singularities and blow-up:** if the true solution becomes infinite, as $\tan x$ does at $\pi/2$, no method can step past it; adaptive solvers will shrink h towards zero.
- **Order assumes smoothness:** if f has a jump or a kink, the observed order drops. Place a grid point at the discontinuity.

10. Applications

10.1 Newton's law of cooling

The rate of temperature change is proportional to the difference from the surroundings:

$$\frac{dT}{dt} = -k\,(T - T_a).$$

Problem. A cup of coffee at 90°C is placed in a 20°C room with $k = 0.1$ min^{-1}. (a) What is its temperature after 10 minutes? (b) When does it reach 30°C?

The equation is separable (and linear): $T(t) = T_a + (T_0 - T_a)e^{-kt} = 20 + 70e^{-0.1t}$.

(a) $T(10) = 20 + 70e^{-1} \approx 20 + 25.752 = 45.75$°C.

(b) $20 + 70e^{-0.1t} = 30 \Rightarrow e^{-0.1t} = 1/7 \Rightarrow t = 10\ln 7 \approx 19.46$ min.

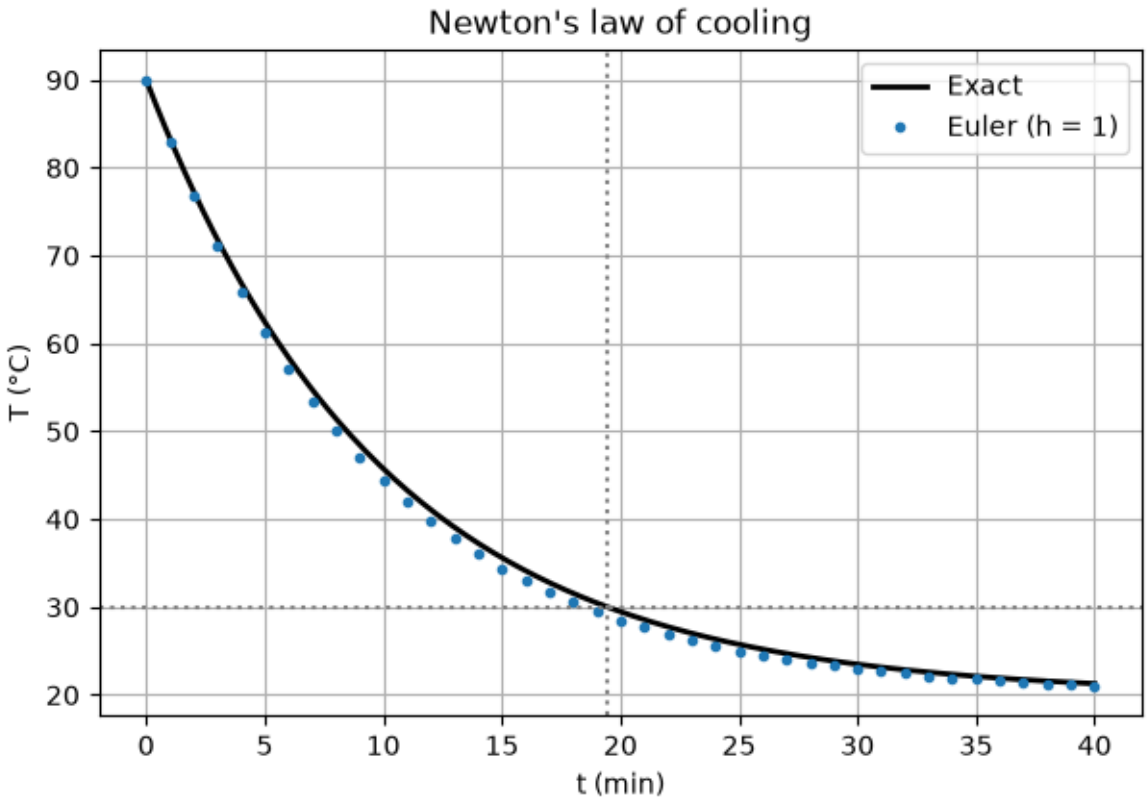
```
k, Ta, T0 = 0.1, 20.0, 90.0
cool = lambda t, T: -k * (T - Ta)
T_exact = lambda t: Ta + (T0 - Ta) * np.exp(-k * t)

t, T_rk = rk4(cool, 0, T0, 1.0, 40)       # h = 1 minute
_, T_eu = euler(cool, 0, T0, 1.0, 40)

print(f"T(10) exact = {T_exact(10):.4f}   RK4 = {T_rk[10]:.4f}   Euler = {T_eu[10]:.4f}")
print(f"time to reach 30 °C = 10 ln 7 = {10 * np.log(7):.4f} min")

tf = np.linspace(0, 40, 200)
plt.plot(tf, T_exact(tf), "k", lw=2, label="Exact")
plt.plot(t, T_eu, ".", label="Euler (h = 1)")
plt.axhline(30, color="gray", ls=":"); plt.axvline(10 * np.log(7), color="gray", ls=":")
plt.xlabel("t (min)"); plt.ylabel("T (°C)"); plt.title("Newton's law of cooling")
plt.legend();
```

```
T(10) exact = 45.7516   RK4 = 45.7516   Euler = 44.4075
time to reach 30 °C = 10 ln 7 = 19.4591 min
```



10.2 Logistic population growth

$$\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right),$$

where r is the growth rate and K is the carrying capacity. Separating variables with partial fractions gives

$$P(t) = \frac{K}{1 + \left(\dfrac{K - P_0}{P_0}\right) e^{-rt}}.$$

Problem. $r = 0.5$ per year, $K = 1000$, $P_0 = 100$. Find $P(5)$.

$$P(5) = \frac{1000}{1 + 9e^{-2.5}} = \frac{1000}{1 + 9(0.082085)} = \frac{1000}{1.738765} \approx 575.1.$$

The equilibria are $P = 0$ (unstable) and $P = K$ (stable). Growth is fastest at $P = K/2$.

```
r, K, P0 = 0.5, 1000.0, 100.0
logistic = lambda t, P: r * P * (1 - P / K)
P_exact = lambda t: K / (1 + (K - P0) / P0 * np.exp(-r * t))

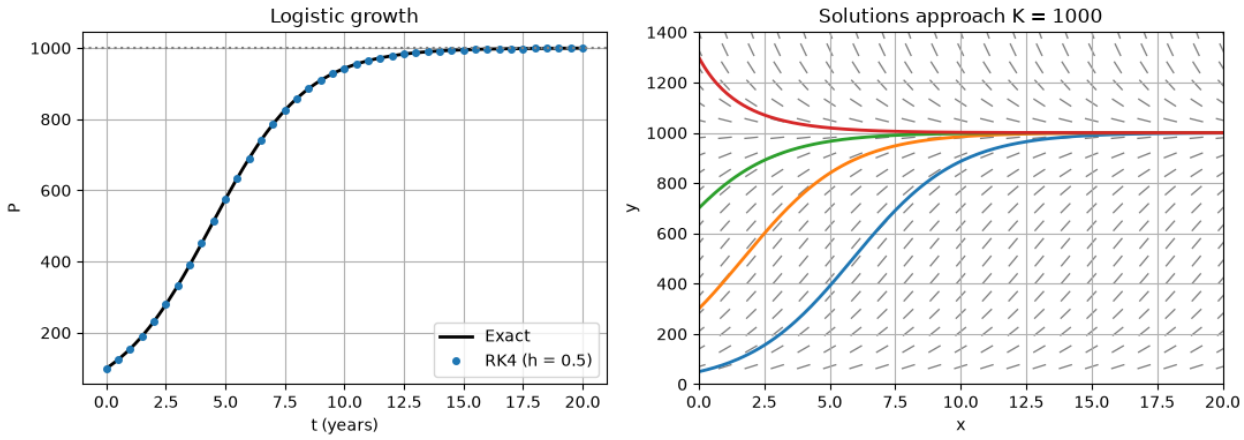
t, P_rk = rk4(logistic, 0, P0, 0.5, 40)
print(f"P(5) exact = {P_exact(5):.4f}   RK4 (h=0.5) = {P_rk[10]:.4f}")

fig, axes = plt.subplots(1, 2, figsize=(11, 4))
tf = np.linspace(0, 20, 200)
axes[0].plot(tf, P_exact(tf), "k", lw=2, label="Exact")
```

```
axes[0].plot(t, P_rk, "o", ms=4, label="RK4 (h = 0.5)")
axes[0].axhline(K, color="gray", ls=":"); axes[0].set_title("Logistic growth")
axes[0].set_xlabel("t (years)"); axes[0].set_ylabel("P"); axes[0].legend()

direction_field(logistic, (0, 20), (0, 1400), ax=axes[1])
for p0 in [50, 300, 700, 1300]:
    _, Pp = rk4(logistic, 0, p0, 0.1, 200)
    axes[1].plot(np.linspace(0, 20, 201), Pp, lw=2)
axes[1].set_title("Solutions approach K = 1000")
plt.tight_layout()
```

```
P(5) exact = 575.1209   RK4 (h=0.5) = 575.1165
```



11. An equation with no elementary solution

The Riccati equation

$$y' = x^2 + y^2, \qquad y(0) = 0$$

cannot be solved in terms of elementary functions. We estimate $y(1)$ with RK4 and check the result by repeatedly halving h. When the digits stop changing, the answer has converged.

```
g = lambda x, y: x**2 + y**2

prev = None
print(f"{'h':>8} {'y(1) RK4':>16} {'change':>12}")
for h in [0.2, 0.1, 0.05, 0.025, 0.0125]:
    _, y = rk4(g, 0, 0, h, round(1 / h))
    change = f"{abs(y[-1] - prev):12.2e}" if prev is not None else ""
    print(f"{h:8.4f} {y[-1]:16.10f} {change}")
    prev = y[-1]
```

```
       h         y(1) RK4       change
  0.2000     0.3502575491
  0.1000     0.3502337418     2.38e-05
  0.0500     0.3502319725     1.77e-06
  0.0250     0.3502318526     1.20e-07
  0.0125     0.3502318448     7.79e-09
```

The last two values agree to about 8 significant figures: $y(1) \approx 0.3502318$. The solution grows quickly afterwards and blows up near $x \approx 2.003$, a reminder that solutions of nonlinear ODEs can exist only on a finite interval.

12. Summary

Type	Form	Method
Separable	$y' = g(x)h(y)$	$\int dy/h = \int g\,dx$
Linear	$y' + Py = Q$	Integrating factor $\mu = e^{\int P}$
Exact	$M\,dx + N\,dy = 0$, $M_y = N_x$	Find F with $F_x = M$, $F_y = N$; then $F = C$
Anything else	$y' = f(x, y)$	Numerical: Euler (order 1), Heun (2), RK4 (4)

13. Exercises

1. Solve $y' = y\cos x$, $y(0) = 2$.
2. Solve $y' - 3y = e^{2x}$, $y(0) = 0$.
3. Show that $(ye^x + 2x)\,dx + (e^x - 1)\,dy = 0$ is exact and solve it with $y(1) = 0$.
4. Use Euler with $h = 0.25$ to estimate $y(1)$ for $y' = -2xy$, $y(0) = 1$. Compare with the exact value e^{-1}.
5. Repeat Exercise 4 with RK4 and $h = 0.25$. How many times smaller is the error?

Answers

1. $y = 2e^{\sin x}$
2. $y = e^{3x} - e^{2x}$
3. $M_y = e^x = N_x$; $F = ye^x + x^2 - y$, so $ye^x + x^2 - y = 1$, giving $y = \dfrac{1 - x^2}{e^x - 1}$.
4. and 5. Run the cell below.

```
q = lambda x, y: -2 * x * y
_, y_eu = euler(q, 0, 1, 0.25, 4)
_, y_rk = rk4(q, 0, 1, 0.25, 4)
ex = np.exp(-1)
print(f"Exact  y(1) = {ex:.6f}")
print(f"Euler  y(1) = {y_eu[-1]:.6f}   error = {abs(y_eu[-1] - ex):.2e}")
print(f"RK4    y(1) = {y_rk[-1]:.6f}   error = {abs(y_rk[-1] - ex):.2e}")
print(f"Error ratio Euler / RK4   {abs(y_eu[-1] - ex) / abs(y_rk[-1] - ex):.0f}")
```

```
Exact  y(1) = 0.367879
Euler  y(1) = 0.410156   error = 4.23e-02
RK4    y(1) = 0.367934   error = 5.51e-05
Error ratio Euler / RK4   768
```